



Orthopedic Exam of the Neck

Dr. Sing

OD3

Thanks to Drs. Lin and Pfothauer who have edited these slides over the years.

Objectives:
After this session
the OMSII will be able to:



Identify additional history-taking elements for neck pain



Perform a MSK focused physical exam, including
indications for provocative tests of the neck.

Resources



Bickley L, Szilagy P. Bates Guide to Physical Exam and History Taking. 13th edition. Chapter 23: The Musculoskeletal System.



Touro MSK MSK exam videos



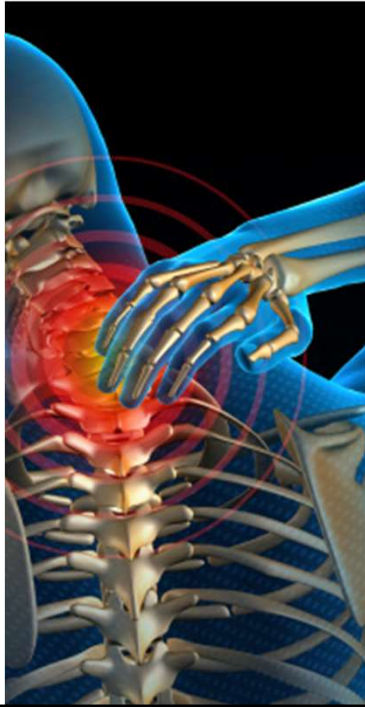
Armstrong A, Hubbard M. Essentials of Musculoskeletal Care. 6th Edition.



Cases:

- A 42-year-old office worker presents with 3 weeks of neck pain after long hours at her desk. Her pain is localized.
 - A 65 yr -old former painter presents with sudden onset neck pain that radiates to his left shoulder and arm for 3 days. He reports some weakness.
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What's on your differential? How do you start?



Neck Pain

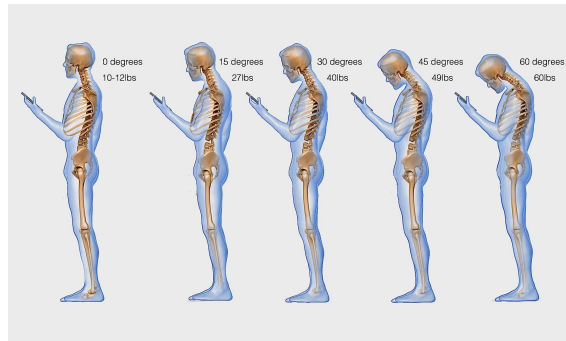
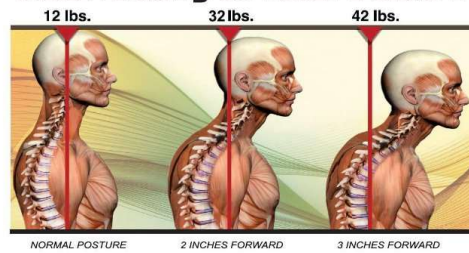
- Commonly seen in primary care up to 20% of visits
- 4th common disability worldwide
- 1 in 5 people who experience neck pain seek medical care.
- Many causes, limited research. Often neck pain is usually self limiting.
- Older adults have degeneration of facet joints and collapse of intervertebral discs.
- Radicular neck pain prevalence– 25 to 45% over age 45
- Nearly 85% of neck pain cases attributed to chronic stress or strains, acute or repetitive trauma injuries (associated with depression, anxiety, poor posture, occupational or sporting injuries)

Neck pain is very common, it is important to have a structured flow to evaluate neck pain in order to recognize diagnoses which require urgent attention.

Am Fam Physician. 2020;102(3):150-156

Neck pain evaluation and initial management

How Heavy is Your Head?



Posture and ergonomics play a significant role in the presentation of neck pain. People are using computers, laptops, on their phones much more than ever before. Look how these everyday posture changes can place stress on the posterior neck and thorax musculature, think OPP and upper crossed syndrome as part of your differential as well!

Google images and Time magazine (pic source)

Differential Diagnosis Overview

Soft Tissue – strain, posture,
whiplash → diffuse aching

Orthopedic – spondylosis, facet
dysfunction → mechanical pain

Neurologic – radiculopathy,
myelopathy → dermatomal pain,
weakness

A. Soft Tissue / Myofascial

- Common: poor posture, strain, “tech neck,” whiplash
- Muscles: upper trapezius, levator scapulae, paraspinals
- Pain is **diffuse, dull, aching**, may radiate to shoulders but not past elbows
- No neurologic deficits**

B. Orthopedic / Structural

- Degenerative joint disease** (cervical spondylosis)
- Facet joint dysfunction**
- Discogenic pain** (with or without radiculopathy)
- Traumatic** (fracture, ligament injury, whiplash)
- Pain may be **localized** or **mechanical**, often worse with motion or loading

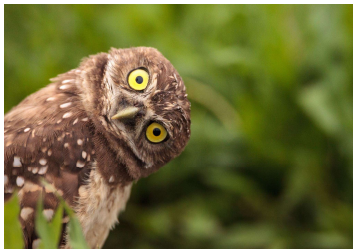
C. Neurologic

- Cervical radiculopathy**: dermatomal pain, numbness, weakness
- Myelopathy**: hand clumsiness, gait instability, hyperreflexia
- Referred pain**: from cardiac, pulmonary, or GI sources (rare but critical)

“

A careful history will lead to the diagnosis 80%
of the time

”



Hampton, BMJ, 1975



Time for oldcarts 2.0, ask more open/specific/clarifying questions

- Neck Pain – obtain a **thorough history**
 - **Was there trauma or injury?**
 - "What were you doing when the pain started?"
 - What happened? How did you land? Was there Loss of consciousness?
 - Was there immediate pain or how long did it take to develop?
 - Were they able to return to work/play/activity?
 - **Exact location of the pain**
 - Specific vertebrae, regional pain, soft tissues of back
 - Localized vs bilateral vs. radiating
- **Duration & timing** – acute, subacute, insidious
- **Character** – ache, dull, stabbing, burning, numbness
- **Comorbid conditions**
- **Radicular symptoms (Radiation)**
- **Aggravating and Relieving Factors**
- **Hx of prior injury/Pain** – include prior work up info and treatments, time frames for the pain
- **PRESENCE OF Weakness**

You are familiar with OLDCARTS to develop questions for the HPI. According to their chief complaint, you can use OLDCARTS but you should ask more in depth clarifying questions to help you figure out if this is truly MSK, neuro, or even rheumatologic.

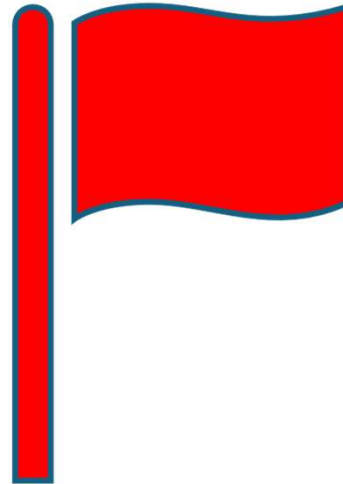
Ask questions so you develop a differential diagnosis and evaluate red flag symptoms.

Here are some questions to incorporate as needed.

Red Flags

These symptoms suggest a worrisome diagnosis ⚠️

- **Infection:** fever, IV drug use, immunosuppression
- **Malignancy:** weight loss, night pain, prior cancer diagnosis
- **Fracture:** major trauma, osteoporosis
- **Neurologic:** new weakness, gait disturbance, hyperreflexia, bowel/bladder change.



MVA OR
TRAUMA:
ASK MORE
QUESTIONS!

- UNDERSTAND THE MECHANISM OF INJURY.
- WHAT HAPPENED? USE OF AIRBAG OR SEATBELT? WAS THERE LOSS OF CONSCIOUSNESS (LOC)?
- DIRECTION OF IMPACT
- SPEED ON IMPACT
- TRAUMA. IMMEDIATE PAIN? SWELLING? POPPING? PAIN WORSE LATER? AMBULANCE?
- PRESENTATION TO ER OR NOT. XRAYS OR OTHER IMAGING PERFORMED?
- IF YOU SUSPECT CERVICAL FRACTURE, XRAY BEFORE PERFORMING PHYSICAL EXAM.

Here are some specific questions you would ask if a patient experiences high level trauma like an MVA or Fall.

X ray or CT scan would be indicated depending on the impact of injury

Physical Exam Overview IPROMS

- INSPECTION
- PALPATION
- RANGE OF MOTION
- STRENGTH
- +/- Neurovascular exam (Reflexes, Dermatomes, pulses)
- Then orthopedic special tests!!



- Examine in an organized and systematic fashion (top down, ladder fashion)
- **INSPECTION** Examine exposed neck, shoulders and upper back (**IPROMS**)
- **PALPATION** OF BONY AND SOFT TISSUES (INCLUDES YOUR OSTEOPATHIC STRUCTURAL EXAMINATION)
- **ASSESS RANGE OF MOTION (ROM)** – ACTIVE AND PASSIVE
- **STRENGTH**
- **NEUROVASCULAR EXAM** (Pulses, Reflexes, dermatomes as indicated)
- **OPP** evaluation (TART, Cervical SD, fascial)
- **SPECIAL Tests**
 - Spurling test
 - Lhermitte
 - Roos
 - Adson Test

I PROMS

Inspection

Palpation

ROM

Special tests

When learning how to perform a comprehensive MSK exam, be systematic, you will often have a working diagnosis before you even get to special tests. Special tests help narrow your differential even further.

Your palpation skills are key in the MSK findings, you will find SD that will help you with your diagnosis or make the diagnosis.



Spurling test

- Assesses for nerve root irritation
- Can be related to spondylitic compression, discogenic compression or stinger-burner syndrome (compression or stretching injury of the brachial plexus)
- To Perform: Examiner extends, side-bends and partially rotates the patient's head to the side being tested. Place an axial load gently to the top of the head
- Positive Test: radiation of pain to the ipsilateral shoulder or arm
- [Spurling Test](#)



[Lhermitte Video](#)



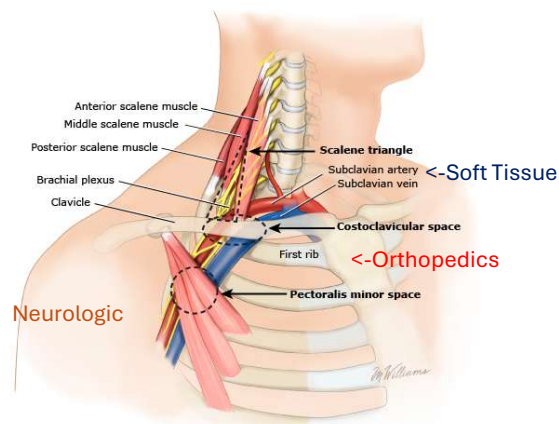
Lhermitte sign

- Assess for cervical radiculopathy
 - Cervical spondylosis, myelopathy, or multiple sclerosis
- To Perform:
 - Forward flexion of the neck
- Positive finding:
 - Paresthesia down spine or back
 - ***Manual cervical distraction may reduce neck and limb symptoms***

Click me!!

Due to hyperexcitability of demyelinating nerves in the spinal cord

Thoracic Outlet Syndrome



- Thoracic outlet syndrome (TOS) is a nonspecific diagnosis representing many conditions that involve the compression of the neurovascular structures that pass through the thoracic outlet.
- The first rib, scalene muscles, and the clavicle comprise the thoracic outlet. But compression can also occur as the neurovascular bundle passes through the scalenes and under the pectoralis minor space.
- Symptoms are varied depending on arterial, nerve, or venous compression.
- Arterial TOS:** People with arterial TOS have an extra rib above the highest rib, closest to the neck. This extra rib compresses the subclavian artery, the blood vessel that carries blood behind the collarbone to your arm and hand. It can cause pain and numbness in the hand due to blood clots. Some people form a dangerous aneurysm, a balloon-like bulge in the artery.
- Neurogenic TOS:** The most common type of TOS, neurogenic refers to nerves. Tears in the scalene muscles on the sides of the neck cause muscle spasms that can irritate the nerves in the brachial plexus.
- Venous TOS:** This type of TOS affects the subclavian vein, which runs between the collarbone and first rib. Athletes who frequently lift or throw a ball overhead

can develop injuries in the vein. Clots and blockages can result

Kaplan J, Kanwal A. Thoracic Outlet Syndrome. [Updated 2023 Apr 10]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK557450/>

Thoracic outlet syndrome

- [Adson Test](#)
 - Perform: extend neck and rotate to affected side and lungs with deep inspiration; palpate the ipsilateral pulse
 - Positive test: decreased amplitude of the radial pulse with this maneuver or reproduction of numbness/tingling per patient.
- [ROOS test](#)



Note: this examination will not diagnose thoracic outlet syndrome but rather make your suspicion very high. You will then order studies to confirm if you suspect vascular or neurogenic.

Best way to confirm if neurogenic is Emg/NCS and vascular is vascular study.

Clinical Pearls & Wrap-Up

- Most neck pain is mechanical & self-limited
- Use history + IPROMS for initial evaluation
- Look for patterns (myofascial vs radicular vs systemic)
- Red flags → imaging or referral



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